

# SS3 HORTICULTURE AND CROP PRODUCTION LESSON NOTES

## FIRST TERM

### WEEK 1: REVIEW OF SS1–SS2 WORK

#### Topic: Questions and answers

#### Introduction

This week is dedicated to refreshing our foundational knowledge from SS1 and SS2. Horticulture is a specialized branch of agriculture, and a strong understanding of the basics is essential before we move into the advanced, high-tech topics of SS3. This review will be interactive, driven by questions and answers.

#### A. Core concepts (recap)

1. **Definition of horticulture:** The art, science, technology, and business of plant cultivation. It is the intensive cultivation of plants for human use, food, and aesthetic pleasure.
2. **Branches of horticulture:**
  - **Pomology:** The science and practice of *fruit* cultivation (e.g., mango, citrus, pawpaw).
  - **Olericulture:** The science and practice of *vegetable* cultivation (e.g., tomato, pepper, leafy greens).
  - **Floriculture:** The science and practice of *flower* and *ornamental plant* cultivation (e.g., rose, hibiscus, carpet grass).
  - **Landscaping/Landscape horticulture:** The art and science of designing, planting, and maintaining aesthetic outdoor spaces (e.g., parks, home gardens, office grounds).
3. **Tools and equipment:** Basic (cutlass, hoe, hand trowel, pruning shears), Intermediate (wheelbarrow, sprayer), and Mechanized (tractor, planter, harvester).
4. **Cultural practices:** These are the essential activities for successful crop production.

- **Land preparation:** Clearing, tilling, and creating beds/ridges.
  - **Propagation methods:**
    - *Sexual:* Using seeds.
    - *Asexual (Vegetative):* Using cuttings, grafting, budding, layering, or suckers.
  - **Planting:** Sowing seeds or transplanting seedlings at the correct spacing and depth.
  - **Weeding:** Controlling unwanted plants that compete for nutrients, water, and light.
  - **Fertilization:** Applying manure (organic) or NPK (inorganic) to supply essential nutrients.
  - **Irrigation:** Supplying water to crops, especially during the dry season.
5. **Pest and disease management:**
- **Pests:** Insects, rodents, birds (e.g., grasshoppers, stem borers, rats).
  - **Diseases:** Caused by fungi, bacteria, or viruses (e.g., leaf spot, blight, mosaic virus).
  - **Control:** Chemical (pesticides, fungicides), Biological (using natural enemies), and Cultural (crop rotation, sanitation).
6. **Harvesting and post-harvest:**
- **Maturity:** Knowing the signs of ripeness.
  - **Handling:** Careful harvesting to prevent bruising.
  - **Storage:** Proper storage (cool, dry place) to reduce spoilage and increase shelf life.

### Evaluation questions

1. Define Olericulture and give three examples of crops it covers.
2. What is the difference between sexual and asexual propagation?
3. List four common cultural practices in horticulture.
4. Differentiate between a pest and a disease.
5. Why is proper post-harvest handling crucial for horticultural crops?

### Assignment questions

1. Draw a table to differentiate between Pomology, Olericulture, and Floriculture.
2. Explain the processes of "grafting" and "budding" in your own words.
3. What are the advantages and disadvantages of using inorganic fertilizer (NPK)?
4. Describe three *non-chemical* methods a farmer can use to control pests.
5. Create a step-by-step "production guide" (from land preparation to harvest) for a common vegetable, such as tomatoes or peppers.



## WEEK 2: ADVANCED GREENHOUSE TECHNOLOGY

### Topic: Climate control, fertigation

#### Introduction

A greenhouse (or "hothouse") is a structure with walls and a roof made of transparent material (glass or plastic) in which plants requiring regulated climatic conditions are grown. While a simple "polytunnel" just traps heat, *advanced greenhouse technology* involves actively managing the environment for maximum yield and quality.

#### A. Climate control systems

The goal is to create the *perfect* micro-climate for a specific crop, regardless of the weather outside.

##### 1. Temperature management:

- **Heating:** Used in cold climates or during cold nights. Can be done with electric heaters, gas heaters, or hot water pipe systems.
- **Ventilation:** Essential for cooling and removing excess humidity. This can be *passive* (roof and side vents that open) or *active* (using large exhaust fans).
- **Evaporative cooling:** A very effective system. Water is pumped over a "pad wall" (looks like cardboard), and fans pull the outside air *through* the wet pad. The water evaporates, significantly cooling the air inside (e.g., "Pad-and-Fan" system).
- **Shade nets:** External or internal nets that can be drawn to reduce sunlight intensity and heat.

##### 2. Humidity control:

- **Misting/Fogging systems:** High-pressure pumps create a fine fog. This raises humidity for sensitive crops (like seedlings) and can also provide a cooling effect.
- **Dehumidifiers:** Used in high-humidity, cool climates where ventilation might bring in *more* moisture, leading to fungal diseases.

##### 3. Light management (Photoperiod control):

- **Supplemental lighting:** Used in temperate regions with short winter days or to "extend the day" to trigger flowering in specific plants. Uses High-Pressure Sodium (HPS) lamps or modern, efficient LEDs (Light Emitting Diodes).
- **Blackout curtains:** Used to "shorten the day" to trigger flowering in plants like chrysanthemums.

#### 4. **Carbon dioxide (CO<sub>2</sub>) enrichment:**

- In a sealed greenhouse, plants can use up the available CO<sub>2</sub>, slowing photosynthesis.
- CO<sub>2</sub> generators (burning propane) or tanks of pure CO<sub>2</sub> are used to raise the CO<sub>2</sub> level (from ~400 ppm to 1000-1500 ppm), which can dramatically boost plant growth.

### **B. Fertigation**

- **Definition:** The combined application of **FERTILizer** and **irriGATION**. It involves injecting liquid fertilizers directly into the irrigation system.
- **Method:**
  1. A "stock solution" of concentrated fertilizer is prepared in a tank.
  2. An "injector" (like a Venturi pump or a Dosatron doser) precisely draws a small amount of the stock solution and mixes it with the main irrigation water.
  3. This nutrient-rich water is delivered to each plant, most commonly through a **drip irrigation system**.
- **Advantages:**
  - **Precision:** Each plant gets the exact amount of nutrients it needs, when it needs it.
  - **Efficiency:** Reduces fertilizer waste (no runoff) and water waste.
  - **Labour saving:** Automates the fertilizing process.
  - **Reduced soil compaction:** No need to walk between rows to apply fertilizer.

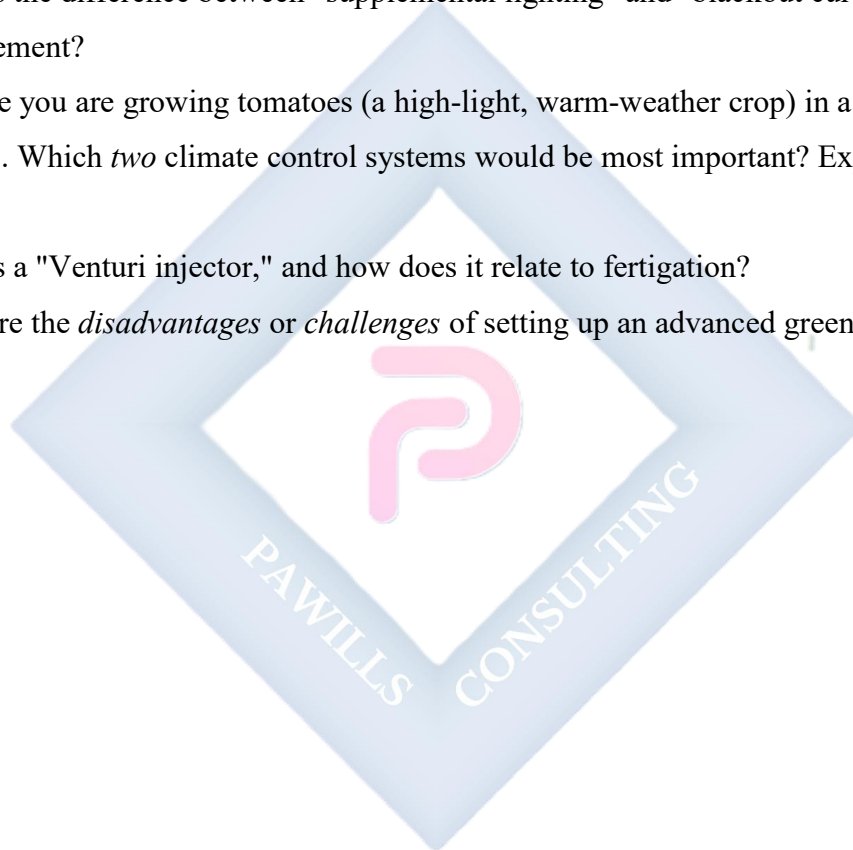
### **Evaluation questions**

1. What is the primary purpose of an advanced greenhouse?
2. Explain how a "pad-and-fan" cooling system works.

3. What is "CO<sub>2</sub> enrichment," and why is it used?
4. Define "fertigation."
5. List three major advantages of using a fertigation system.

### Assignment questions

1. Draw a simple diagram of a greenhouse, and label the following parts: ventilation fan, cooling pad, shade net, and drip irrigation lines.
2. What is the difference between "supplemental lighting" and "blackout curtains" in light management?
3. Imagine you are growing tomatoes (a high-light, warm-weather crop) in a hot, dry climate. Which *two* climate control systems would be most important? Explain your choice.
4. What is a "Venturi injector," and how does it relate to fertigation?
5. What are the *disadvantages* or *challenges* of setting up an advanced greenhouse?



## WEEK 3: HYDROPONICS

### Topic: Meaning, types, advantages

#### Introduction

Hydroponics is a method of growing plants in a soilless medium, or with no medium at all, using mineral nutrient solutions dissolved in water. It is a high-tech, intensive method of cultivation.

#### A. Meaning and core principles

- The word comes from Greek: hydro (water) and ponos (labour).
- **It is not just "growing plants in water."** It is growing plants in a *highly controlled nutrient solution* that provides all 13 essential macro- and micro-nutrients.
- The roots must also have access to **oxygen**. This is a critical component.

#### B. Advantages and disadvantages

- **Advantages:**
  - **Water conservation:** Uses up to 90% less water than traditional farming, as water is recirculated.
  - **No soil-borne pests/diseases:** Eliminates problems like nematodes, fungi, and weeds.
  - **Faster growth:** Plants get the perfect amount of food and oxygen, so they grow faster.
  - **Higher yields:** More plants can be grown in a smaller space (vertical farming).
  - **Total control:** The grower has 100% control over the nutrient mix.
  - **Grow anywhere:** Can be set up indoors, on rooftops, in deserts, etc.
- **Disadvantages:**
  - **High initial cost:** Pumps, tanks, and systems are expensive.
  - **Technical skill:** Requires knowledge of chemistry and plant nutrition.
  - **Power dependent:** Most systems are "active" and will fail if the electricity goes out.

- **Rapid disease spread:** If a water-borne disease (like Pythium) gets in, it can wipe out the entire system quickly.

### C. Types of hydroponic systems

Systems are either *Passive* (no pumps) or *Active* (use pumps and timers).

1. **Wick system (Passive):** The simplest method. A nylon wick draws the nutrient solution from a reservoir up into a soilless growing medium (like coco coir or perlite) where the roots are.
2. **Deep water culture (DWC) (Active):**
  - Plants are held in floating styrofoam rafts or net pots, with their roots suspended directly in a tank of nutrient solution.
  - An **air pump** with an **air stone** (like in a fish tank) constantly bubbles oxygen into the water.
  - Excellent for fast-growing leafy greens like lettuce.
3. **Nutrient film technique (NFT) (Active):**
  - Plants are grown in shallow, sloped channels or gullies.
  - A continuous, *thin film* of nutrient solution flows down the channel, over the bare roots.
  - The film provides nutrients, and the ample air space in the gully provides oxygen.
4. **Ebb and flow (Flood and drain) (Active):**
  - Plants are in a "grow bed" filled with a sterile medium (like hydroton/clay pebbles).
  - A pump (on a timer) *floods* the grow bed with the nutrient solution from a reservoir.
  - When the timer turns off, the solution *drains* (ebbs) back into the reservoir, pulling fresh oxygen down to the roots.
5. **Drip system (Active):**
  - Similar to fertigation. Drip emitters deliver the nutrient solution directly to the base of each plant in a soilless medium (e.g., rockwool, coco coir).



- Can be "recovery" (unused solution drains back to the reservoir) or "non-recovery" (excess drains away).

### **Evaluation questions**

1. What is hydroponics?
2. Besides water and nutrients, what is the third critical element that plant roots need in a hydroponic system?
3. List three major advantages of hydroponics over soil farming.
4. List three disadvantages.
5. What is the main difference between a DWC system and an NFT system?

### **Assignment questions**

1. Draw a simple, labelled diagram of a Deep Water Culture (DWC) system. (Must include: tank, raft, net pot, air pump, air stone).
2. Draw a simple, labelled diagram of a Nutrient Film Technique (NFT) system. (Must include: reservoir, pump, sloped channel, plant).
3. What is the difference between a "passive" and an "active" hydroponic system? Give one example of each.
4. If your electricity supply is very unreliable, which *two* hydroponic systems would be the *riskiest* to run? Why?
5. What is a "soilless medium" or "substrate"? List three examples of materials used.

## WEEK 4: AEROPONICS

### Topic: Meaning, method, advantages

#### Introduction

Aeroponics is the most advanced form of hydroponics. It is the process of growing plants in an air or mist environment, *without the use of soil or an aggregate medium*.

#### A. Meaning

- From the Greek: *aero* (air) and *ponos* (labour).
- Plant roots are suspended in a dark, enclosed chamber.
- They are intermittently (or continuously) misted with a fine, nutrient-rich fog.
- This method provides the *maximum* possible oxygen to the root zone, leading to extremely fast growth.

#### B. Method (How it works)

1. **Chamber:** Plants are placed in supports (plastic clips, foam collars) at the top of a chamber (a box, tube, or bucket). The stems are up in the light, and the roots hang down in the dark.
2. **Misting system:** A nutrient solution is stored in a reservoir.
  - **High-Pressure Aeroponics (HPA):** A high-pressure pump (60-100 psi) forces the solution through very fine *mister nozzles*, creating a true fog (droplets of 50 microns or less) that can be absorbed efficiently by root hairs.
  - **Low-Pressure Aeroponics (LPA):** Uses a simple pond pump and sprinkler heads. This creates more of a "spray" than a "mist." It's cheaper but less effective.
3. **Timer:** The pump is connected to a high-precision timer. It is programmed to mist for a very short duration, followed by a longer pause (e.g., 3-5 seconds of mist every 10-15 minutes). This keeps the roots moist but never saturated, allowing for constant oxygen uptake.

#### C. Advantages

1. **Maximum oxygen exposure:** This is the biggest advantage. More oxygen at the root zone = faster nutrient uptake = faster growth.
2. **Fastest growth rates:** Aeroponics is generally considered the fastest-growing method.
3. **Water and nutrient efficiency:** Uses even *less* water and nutrients than hydroponics, as nothing is wasted.
4. **Mobility:** Systems are very lightweight (no soil, no water).
5. **Easy root inspection:** You can easily open the chamber to check root health for pests or disease.

#### D. Disadvantages

1. **Extreme power dependency:** This is its main weakness. If the pump fails or electricity is cut for even 30-60 minutes, the fine root hairs can dry out and the entire crop can die.
2. **High cost:** High-pressure pumps and fine mister nozzles are expensive.
3. **Clogging:** The tiny nozzles can easily clog with mineral salts, requiring frequent maintenance.
4. **High technical skill:** Requires the most skill to manage.

#### Evaluation questions

1. What is aeroponics?
2. What is the *primary* difference between aeroponics and deep water culture (DWC)?
3. List two advantages of aeroponics.
4. What is the *biggest* risk or disadvantage of running an aeroponic system?
5. What is the difference between HPA (High-Pressure Aeroponics) and LPA (Low-Pressure Aeroponics)?

#### Assignment questions

1. Draw a simple, labelled diagram of an aeroponic system. (Must include: chamber, plant support, hanging roots, mister nozzle, reservoir, pump).
2. Explain why an aeroponic system is so dependent on a timer.

3. If you were comparing NFT, DWC, and Aeroponics, which would you say provides the most oxygen to the roots? Why?
4. Why must the root chamber in an aeroponic system be dark?
5. A farmer's aeroponic nozzles keep clogging. What is the most likely cause, and what maintenance should they perform?



## WEEK 5: MIDTERM EXAMINATION

### Topic: Examination

This week is for the midterm test, which will cover all content from Weeks 1-4.

- **Topics:** Review of SS1/SS2, Advanced Greenhouse Technology (climate control, fertigation), Hydroponics (types, pros/cons), and Aeroponics (method, pros/cons).
- **Format:** A combination of objective questions, short-answer questions, and diagram interpretation.



## WEEK 6: AQUAPONICS

### Topic: Integration of fish and crop production

#### Introduction

Aquaponics is a sustainable, closed-loop food production system that combines **AQUA**culture (raising aquatic animals like fish) with hydro**PONICS** (growing plants in water). It creates a symbiotic relationship where both the fish and the plants benefit.

#### A. The aquaponic cycle (The nitrogen cycle)

This is the heart of the system.

1. **Step 1: Fish waste:** Fish are fed. They excrete waste (faeces and urine) that is rich in **Ammonia (NH<sub>3</sub>)**, which is toxic to them.
2. **Step 2: Bacterial conversion:** Water from the fish tank is pumped to the plant "grow bed." This grow bed (filled with gravel, clay pebbles, or just rafts) is home to a colony of beneficial *nitrifying bacteria*.
  - First, bacteria (like *Nitrosomonas*) convert the Ammonia (NH<sub>3</sub>) into **Nitrites (NO<sub>2</sub>)**.
  - Second, other bacteria (like *Nitrobacter*) convert the Nitrites (NO<sub>2</sub>) into **Nitrates (NO<sub>3</sub>)**.
3. **Step 3: Plant uptake:** Nitrates (NO<sub>3</sub>) are a perfect, readily available fertilizer for plants. The plant roots absorb these nitrates, along with other nutrients from the fish waste.
4. **Step 4: Clean water returns:** By absorbing the nitrates, the plants act as a natural, biological filter. The now-clean, oxygenated water flows back to the fish tank, keeping the fish healthy.

#### B. The three living components

1. **The fish:** (e.g., Tilapia, Catfish). They are the "engine" of the system, producing the fertilizer.

2. **The plants:** (e.g., Lettuce, kale, tomatoes, peppers). They are the "filter," cleaning the water.
3. **The bacteria:** (The invisible workers). They are the "bridge" that converts the fish waste into plant food. Without them, the system collapses.

### C. System types

Aquaponic systems use the same designs as hydroponics, with the fish tank replacing the nutrient reservoir.

- **Media-based system:** A grow bed is filled with gravel or clay pebbles. It is "flooded and drained" (Ebb and Flow) using an auto-siphon. This media provides a huge surface area for the bacteria.
- **Deep water culture (DWC):** Plants float on rafts directly in the water that flows from the fish tank.
- **Nutrient film technique (NFT):** Only suitable for small systems, as the fish waste can clog the narrow channels.

### D. Advantages

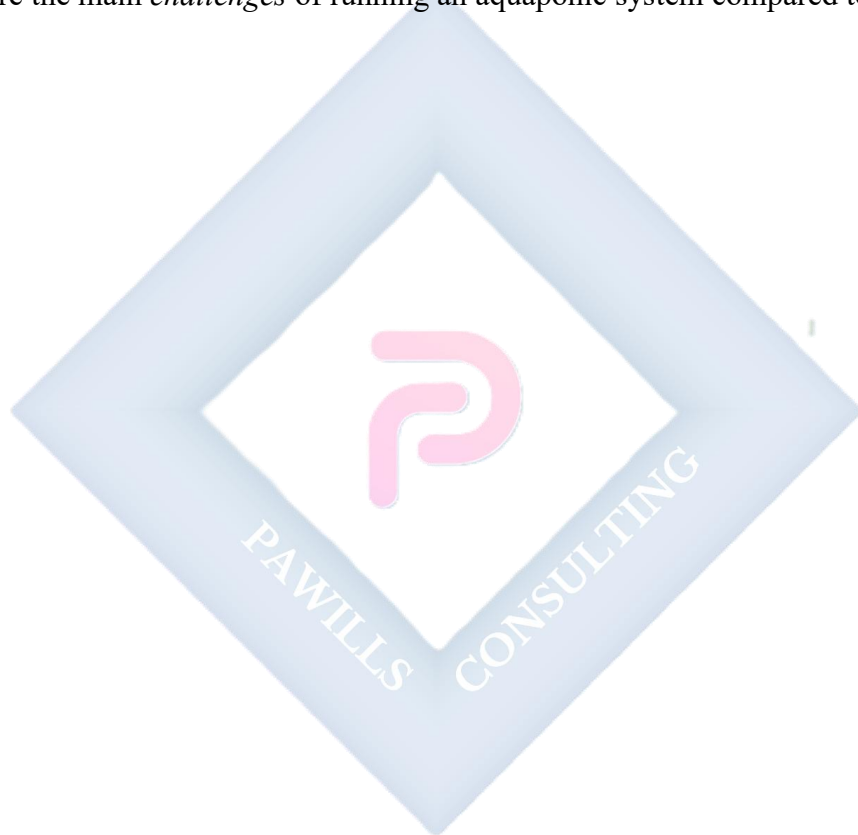
- **Sustainability:** A closed-loop system. Waste is a resource.
- **Water conservation:** Uses ~90% less water than traditional farming.
- **Dual income:** The farmer can harvest and sell both fish *and* plants.
- **Organic:** No need for artificial fertilizers, as the fish provide them.
- **No soil pests.**

### Evaluation questions

1. What two practices are combined to create "aquaponics"?
2. What are the *three* essential living components of an aquaponic system?
3. What toxic substance do fish produce, and what do plants absorb?
4. What is the job of the *bacteria* in the system?
5. What is a "media-based" aquaponic system?

### Assignment questions

1. Draw a simple, labelled diagram of the aquaponic cycle. (Must show: fish tank, pump, grow bed, plants, and the flow of water).
2. What is the difference between an aquaponic system and a hydroponic system? (Hint: Where do the nutrients come from?)
3. What is an "auto-siphon," and which aquaponic system type uses it?
4. What are two common fish species used in aquaponics in Nigeria?
5. What are the main *challenges* of running an aquaponic system compared to hydroponics?





**WEEK 7: MIDTERM BREAK**



## WEEK 8: PRECISION AGRICULTURE

### Topic: ICT tools, drones, sensors

#### Introduction

Precision Agriculture (or "Precision Farming") is a farm management concept that uses Information and Communication Technology (ICT) to make farming more accurate and efficient. Instead of treating a whole field as one unit, it manages specific zones (or even individual plants) differently, based on real-time data.

#### A. The core concept

- **Observe:** Use technology to collect data about the field.
- **Measure:** Analyse the data to find variations (e.g., one part of the field is dry, another part is fertile).
- **Respond:** Use technology to apply *exactly* what is needed, *exactly* where it is needed.

#### B. Key technologies and ICT tools

##### 1. GPS (Global Positioning System):

- **Auto-steer:** Tractors equipped with high-precision GPS can drive themselves in perfectly straight lines, reducing overlap and saving fuel, seeds, and fertilizer.
- **Yield mapping:** A GPS on a harvester maps the yield as it is harvested, showing which parts of the field are most (or least) productive.

##### 2. Sensors:

- **Soil sensors:** Can be driven through a field or placed permanently to measure soil moisture, pH, compaction, and nutrient levels (N, P, K).
- **Crop sensors:** (e.g., Greenseeker). These sensors shine light on plants and read the reflection (NDVI - Normalized Difference Vegetation Index) to measure plant health and "greenness."

##### 3. Drones (UAVs - Unmanned Aerial Vehicles):

- **Scouting:** A farmer can fly a drone over a large field in minutes to check for pest attacks, disease outbreaks, or irrigation leaks.
- **Multispectral imaging:** Drones with special cameras can create maps that show plant stress before it is visible to the human eye.
- **Spot spraying:** Some advanced drones are equipped with small tanks to spray pesticides *only* on the small patch of weeds or pests they detect.

#### 4. VRT (Variable Rate Technology):

- This is the "response" part.
- A prescription map (created from the sensor/drone data) is loaded into a tractor's computer.
- The computer tells the seeder, fertilizer spreader, or sprayer to *vary* its application rate automatically as it drives across the field.
- **Result:** More seeds are applied in low-yield areas, and less fertilizer is applied in already-fertile areas.

#### Evaluation questions

1. What is the main idea behind "Precision Agriculture"?
2. What does "GPS" stand for, and name one way it is used in farming.
3. What is an "NDVI," and what does it measure?
4. List two uses of drones (UAVs) in horticulture.
5. What is "Variable Rate Technology (VRT)"?

#### Assignment questions

1. How does Precision Agriculture help make farming more "sustainable" or "environmentally friendly"?
2. What is the difference between "whole-field management" and "precision agriculture"?
3. Imagine you have a 10-hectare maize farm. Explain how you would use a *drone* and a *VRT fertilizer spreader* to save money and increase yield.
4. What is "yield mapping," and how does it help a farmer make better decisions for the *next* season?

5. What are the main *barriers* or *challenges* to adopting precision agriculture for small-scale farmers in Nigeria?



## WEEK 9: CLIMATE CHANGE AND CROP PRODUCTION

### Topic: Adaptation strategies

#### Introduction

Climate change refers to long-term shifts in temperatures and weather patterns. For horticulture, this is a major threat, causing heat stress, unpredictable rainfall (droughts and floods), and new pest/disease pressures. This week, we discuss how farmers can *adapt*.

#### A. Major impacts of climate change on horticulture

1. **Temperature stress:** Higher average temperatures can "scorch" fruits, reduce flowering, and slow the growth of cool-weather crops (like lettuce).
2. **Erratic rainfall:**
  - **Droughts:** Longer and more severe dry seasons kill plants and reduce yields.
  - **Floods:** Intense, heavy rainfall can erode soil, waterlog roots (causing rot), and wash away crops.
3. **Pest and disease patterns:** Warmer winters allow pests to survive longer. New pests and diseases may migrate to areas that were previously too cold for them.
4. **Salinization:** In coastal areas, rising sea levels can cause salt water to intrude into fresh water sources, making irrigation water and soil too salty for most crops.

#### B. Adaptation strategies (How farmers can respond)

"Adaptation" means adjusting to the new reality to reduce vulnerability.

1. **Genetic strategies:**
  - **Use of resilient varieties:** Farmers can plant crop varieties that have been bred to be *drought-tolerant*, *heat-tolerant*, or *salt-tolerant*.
  - **Crop diversification:** Planting a *variety* of different crops (e.g., maize, beans, and cassava) instead of just one (monoculture). If one crop fails due to drought, the others may survive.
2. **Agronomic strategies:**

- **Changing planting dates:** Shifting planting to an earlier or later time to better match the new, unpredictable rain patterns.
  - **Water conservation techniques:**
    - **Mulching:** Covering the soil with straw, plastic, or wood chips to reduce water evaporation.
    - **Drip irrigation:** Applying water directly to the plant roots (highly efficient).
    - **Rainwater harvesting:** Collecting and storing rainwater from rooftops or in small farm ponds.
  - **Agroforestry:** Integrating trees with crops. The trees provide shade (cooling the crops), reduce soil erosion, and can provide a second source of income (e.g., planting cocoa under shade trees).
3. **Technological strategies:**
- **Protected cultivation:** Using greenhouses and shade nets (from Week 2) to control the environment and protect crops from extreme weather.
  - **Soilless culture:** Using hydroponics or aquaponics (from Week 3-6) to grow food without relying on depleted soil or unpredictable rain.

### Evaluation questions

1. List three major impacts of climate change on crop production.
2. What is an "adaptation strategy"?
3. Explain "crop diversification" and why it is a good adaptation strategy.
4. What is "mulching," and how does it help a farmer adapt to drought?
5. How is agroforestry a good strategy for dealing with high temperatures?

### Assignment questions

1. Differentiate between "climate change mitigation" (stopping the cause) and "climate change adaptation" (living with the effects).
2. What is "rainwater harvesting"? Describe one way a small-scale farmer could do this.

3. How can greenhouse technology (Week 2) be used as a climate change adaptation strategy?
4. What is a "drought-tolerant" crop variety? Research and name one example.
5. Write a short advisory (1-2 paragraphs) to a local farmer in your area, suggesting three simple and low-cost ways they can adapt to the changing weather patterns.



## WEEK 10: SUSTAINABLE HORTICULTURAL PRACTICES

### Topic: Organic farming, permaculture

#### Introduction

"Sustainable" means to produce food in a way that is environmentally sound, economically viable, and socially responsible. It is about meeting our needs today without damaging the ability of future generations to meet their own. This week, we focus on two key approaches: organic farming and permaculture.

#### A. Organic farming

- **Definition:** A production system that sustains the health of soils, ecosystems, and people. It relies on ecological processes and cycles *instead of* synthetic (man-made) inputs.
- **Core Principles (What it *avoids*):**
  - No synthetic pesticides.
  - No synthetic fertilizers (like NPK).
  - No Genetically Modified Organisms (GMOs).
- **Core Principles (What it *uses*):**
  - **Soil health:** This is the #1 priority. Organic farmers "feed the soil, not the plant." This is done with:
    - *Compost:* Decomposed organic matter.
    - *Manure:* Animal waste.
    - *Cover crops / Green manure:* Plants like beans or clover that are grown and then tilled into the soil to add nitrogen.
  - **Pest control:**
    - *Biological:* Encouraging natural predators (e.g., ladybugs, birds).
    - *Cultural:* Crop rotation, intercropping (planting different crops together).
  - **Weed control:** Mulching, hand-weeding, or flame weeders.

#### B. Permaculture ("Permanent Agriculture")



- **Definition:** Permaculture is much broader than organic farming. It is a *design philosophy* for creating sustainable human habitats. It mimics the patterns and relationships found in nature to design landscapes that produce food, fibre, and energy with minimal effort and waste.
- **Core Ethics:**
  1. **Care for the Earth:** (Protect soil, forests, and water).
  2. **Care for the People:** (Ensure access to food, shelter, and health).
  3. **Fair Share:** (Share surplus, limit consumption).
- **Key Design Principles:**
  - **Observe and interact:** Watch the land, wind, and water *before* you design.
  - **Catch and store energy:** (e.g., building a pond to store rainwater, a greenhouse to store solar heat).
  - **Use and value diversity:** Plant many different types of crops together in a "polyculture," not a "monoculture."
  - **Use edges:** The "edge" between two ecosystems (e.g., where a pond meets the land) is the most productive zone.
  - **Integrate, don't segregate:** Each element should serve many functions. (e.g., a chicken tractor: the chickens lay eggs, eat pests, till the soil, and add manure all at once).

### Evaluation questions

1. What is the main goal of "sustainable" horticulture?
2. Define organic farming.
3. What is the *number one priority* for an organic farmer?
4. What is permaculture?
5. What are the three core ethics of permaculture?

### Assignment questions

1. What is the main difference between "organic farming" (a set of rules) and "permaculture" (a design philosophy)?

2. What is a "cover crop"? Give an example and explain how it improves the soil.
3. What is a "polyculture," and why is it more resilient than a "monoculture"?
4. Describe what a "chicken tractor" is and how it demonstrates the permaculture principle of "integration."
5. Design a simple "Permaculture Zone 1" garden for a home backyard, including at least 3 elements (e.g., kitchen garden, compost pile, small rainwater barrel).



## **WEEK 11-13: REVISION & EXAMINATION**

- **WEEK 11:** Comprehensive review of all topics from the term. Focus on the high-tech systems (Greenhouse, Hydroponics, Aeroponics, Aquaponics) and the modern management concepts (Precision Ag, Sustainability).
- **WEEK 12:** Final Term Examination.
- **WEEK 13:** Closing and review of exam scripts.



## SS3 HORTICULTURE AND CROP PRODUCTION LESSON NOTES

### SECOND TERM

#### WEEK 1: EXPORT CROPS IN NIGERIA

#### TOPIC: COCOA, CASHEW, KOLA NUT

##### Introduction

Export crops, also known as "cash crops," are agricultural commodities grown specifically for export to other countries, rather than for local consumption. They are a critical source of foreign exchange (foreign currency) for Nigeria, second only to crude oil. This week, we will study the "big three" horticultural export crops.

##### A. Cocoa (*Theobroma cacao*)

- **Economic importance:** Historically Nigeria's most important non-oil export. It is the primary raw material for chocolate, cocoa powder, and cocoa butter (cosmetics).
- **Ecology:** A tropical forest crop. It requires:
  - **Climate:** High rainfall (1500-2000mm), high humidity (70-80%), and high temperature (25-30°C).
  - **Soil:** Deep, fertile, well-drained loamy soil.
  - **Shade:** Young cocoa plants require shade, so they are often grown under larger trees like plantain, banana, or forest trees.
- **Major production states:** Ondo, Cross River, Ogun, Osun, Edo, Ekiti.
- **Key cultivation practices:** Grown from seeds (in nurseries) or seedlings. Requires pruning and regular weeding.
- **Processing for export:** This is the most critical part.
  1. **Harvesting:** Pods are cut from the tree when ripe (yellow/orange).
  2. **Breaking:** Pods are broken open with a blunt cutlass or wooden club (never a sharp knife) to extract the wet, pulp-covered beans.

3. **Fermentation:** The beans are piled in a heap on banana leaves or in wooden "sweat-boxes" for 5-7 days. This is *essential*. Bacteria and yeast develop, which kills the bean's embryo, breaks down the pulp, and develops the characteristic chocolate flavour and brown colour.

4. **Drying:** The fermented beans are spread thinly on mats or concrete floors and sun-dried for 1-2 weeks until the moisture content is 6-7%.

- **Pests and diseases:**

- **Black pod disease:** A fungus (*Phytophthora*) that causes pods to rot and turn black.
- **Swollen shoot virus:** A viral disease (no cure) that causes a "swollen" appearance on new stems and reduces yield.
- **Capsids (Mirids):** Insects that suck sap from young shoots and pods, causing them to die.

## **B. Cashew (*Anacardium occidentale*)**

- **Economic importance:** A rapidly growing export. Nigeria is one of the world's top producers of raw cashew nuts.
- **Products:** It yields two main products:
  1. **The cashew nut:** This is the primary export product, found at the end of the "apple."
  2. **The cashew apple:** A fleshy, "pseudo-fruit" (false fruit) that is high in Vitamin C. It is eaten fresh or processed into juice, jam, and "cashew wine."
- **Ecology:** A very hardy, drought-resistant tree. It thrives in the savanna and transitional zones where other crops like cocoa cannot.
- **Major production states:** Kogi, Benue, Enugu, Kwara, Ogun, Osun.
- **Processing for export:** Very complex due to the **Cashew Nut Shell Liquid (CNSL)**, a toxic, caustic liquid found in the shell.
  1. **Harvesting:** The apples (with nuts attached) fall to the ground when ripe.
  2. **Roasting/Steaming:** The raw nuts must be heated. This makes the shell brittle and neutralizes the CNSL.
  3. **Shelling:** The nuts are cracked open to release the kernel.

4. **Drying & Peeling:** The kernel has a thin skin (testa) that is removed by drying and peeling. The final white kernel is what is exported.

### C. Kola Nut (*Cola* spp.)

- **Economic importance:** A major cash crop, both for internal trade (especially to Northern Nigeria) and for export. It is used as a stimulant (contains caffeine) and is culturally significant in ceremonies.
- **Species:**
  - ***Cola nitida*:** The most common commercial species. Has two cotyledons (seed leaves).
  - ***Cola acuminata*:** Has more than two cotyledons (3-6). More common in traditional settings in the East.
- **Ecology:** A forest crop, similar to cocoa.
- **Processing for export:**
  1. **Harvesting:** Pods are harvested from the tree.
  2. **"Sweating":** The pods are kept in a heap for a few days to soften (ferment) the outer husk.
  3. **Extraction:** The pods are broken, and the nuts are removed.
  4. **Washing & Curing:** The nuts are washed and cured (a slow drying process) to prepare them for storage. They are stored in baskets lined with leaves to keep them moist.
  - **Pests:** The **kola weevil** is a major storage pest that bores holes in the nuts.

### Evaluation questions

1. What is the primary economic importance of export crops?
2. Why is "fermentation" a necessary step in cocoa processing?
3. Name the two main products of the cashew tree.
4. Why is processing cashew nuts considered dangerous or difficult?
5. What is the main difference between *Cola nitida* and *Cola acuminata*?

### Assignment questions

1. Describe the full processing of cocoa, from harvesting the pod to the final dried bean.
2. What is the "Black Pod Disease" of cocoa? (What is it, what causes it, and how does it affect the farm?)
3. What is CNSL, and where is it found?
4. Explain the ecological (climate, soil) requirements for growing cocoa.
5. Why is the "kola weevil" a major problem for kola nut farmers?



## WEEK 2: GLOBAL TRENDS IN HORTICULTURE

### TOPIC: ORGANIC CERTIFICATION, AGRIBUSINESS

#### Introduction

The world of horticulture is changing. It is no longer just about subsistence farming; it is a high-tech, global industry. This week, we explore the major trends shaping this industry, focusing on the concepts of "Agribusiness" and "Organic Certification."

#### A. Agribusiness

- **Definition:** Agribusiness is the sum of all activities involved in the food and fibre industry. It is not just "farming." It is the *business* of agriculture.
- **Scope:** Agribusiness includes:
  1. **Input supply:** Companies that make and sell seeds, fertilizers, chemicals, and machinery.
  2. **Farm production:** The actual growing of crops (the farm itself).
  3. **Post-harvest handling:** Storage, grading, and packing.
  4. **Processing (Value Addition):** Turning raw crops into other products (e.g., tomato into paste, mango into juice).
  5. **Marketing and distribution:** Transport, advertising, logistics, and sales.
- **The Trend:** The global trend is to view farming as a business (agribusiness), not just a way of life. This means focusing on profit, efficiency, and market demand.

#### B. Global trends in consumer demand

Consumers in wealthy nations (Europe, North America, Asia) are the main buyers of exported horticultural products. Their demands are changing:

1. **Food safety and traceability:** Consumers want to know their food is safe (no harmful pesticide residues) and where it came from (traceability).
2. **Sustainability:** Consumers prefer food grown in a way that does not harm the environment.



3. **Ethical production:** Consumers are concerned about "Fair Trade" (farmers getting paid a fair price) and labour practices (no child labour).

### C. Organic certification

- **What is "Organic"?** Organic farming is a system (from Term 1, Week 10) that avoids synthetic fertilizers, pesticides, and GMOs, focusing instead on building soil health.
- **What is "Certification"?** This is the key concept. "Certification" is a formal, legal process of verification. A third-party "certifying agency" (e.g., Ecocert, Soil Association) inspects a farm to *prove* that it is following all the rules of organic farming.
- **The certification process:**
  1. **Application:** The farmer applies to a certifier.
  2. **Conversion period:** The farmer must farm organically for a "conversion period" (usually 2-3 years) to "cleanse" the land of old chemicals.
  3. **Inspection:** An inspector visits the farm, checks all records (what seeds were bought, what was used for pests), and takes soil/leaf samples.
  4. **Approval:** If the farm passes, it receives an "Organic Certificate" and can legally use the "Organic" logo.
  5. **Annual review:** The inspection is repeated every year.
- **Why is certification necessary?**
  - **Trust:** It's a guarantee for the consumer. Anyone can *claim* to be organic, but the certificate *proves* it.
  - **Market access:** Most international markets will *not* allow you to sell a product as "organic" without a valid certificate.
  - **Premium price:** Certified organic products can be sold for a 20-100% higher price than conventional products.

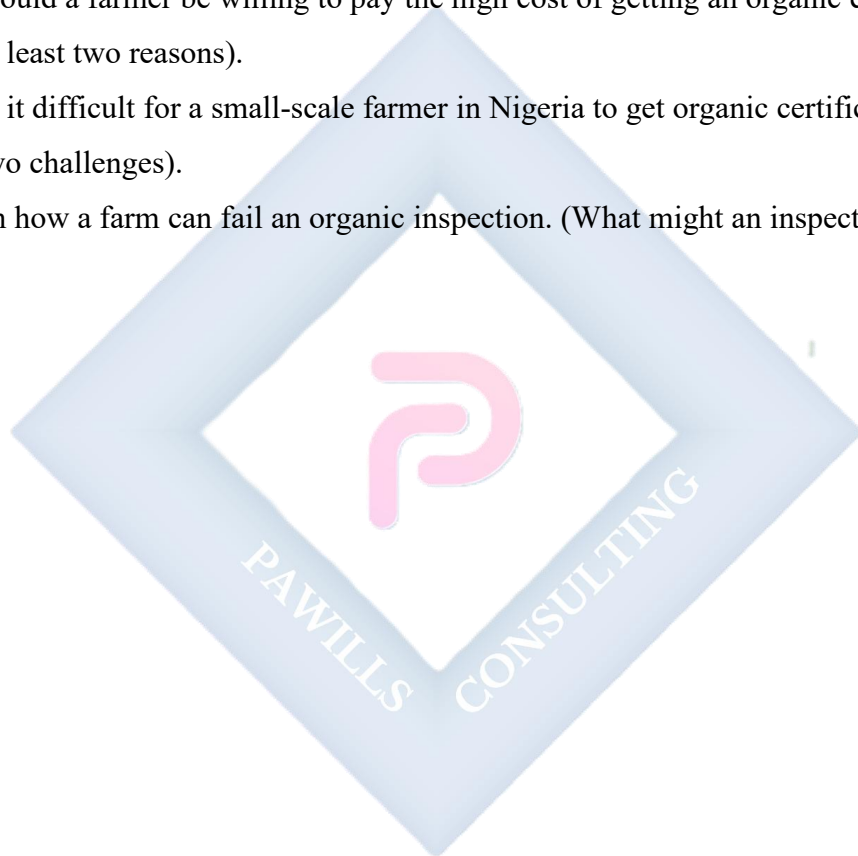
### Evaluation questions

1. Define "Agribusiness" in your own words.
2. List the five main components of the agribusiness sector.
3. What is "organic certification"?

4. Why is a "conversion period" necessary before a farm can be certified organic?
5. What is the main benefit of "traceability" for a consumer?

### **Assignment questions**

1. What is the difference between "organic farming" (a practice) and "organic certification" (a legal process)?
2. List three global trends that are shaping consumer demand for horticultural products.
3. Why would a farmer be willing to pay the high cost of getting an organic certificate? (List at least two reasons).
4. Why is it difficult for a small-scale farmer in Nigeria to get organic certification? (List at least two challenges).
5. Explain how a farm can fail an organic inspection. (What might an inspector find?)



## WEEK 3: AGRIBUSINESS OPPORTUNITIES

### TOPIC: VALUE CHAIN IN HORTICULTURE

#### Introduction

The "value chain" describes the full range of activities needed to bring a product from its initial conception (e.g., the seed) to the final consumer. At each step, "value" is added to the product. The biggest opportunities in modern agribusiness are not just in farming, but in "moving up the value chain."

#### A. Understanding the value chain

A raw product (like a fresh mango) is cheap. A processed product (like packaged, dried mango) is expensive. The "value" is the difference.

- **Farm Gate Price:** The low price a farmer gets when selling from the farm.
- **Retail Price:** The high price the consumer pays in a supermarket.
- **Value Chain:** The chain of activities (processing, packaging, transport) that adds this value.

#### B. Stages of the horticultural value chain & opportunities

##### 1. INPUT SUPPLY:

- **Activities:** Selling seeds, fertilizers, pesticides, irrigation equipment, greenhouses, and tools.
- **Opportunity:** Become a certified distributor for high-quality, improved seeds (e.g., hybrid tomatoes) or build and sell simple, low-cost drip irrigation kits.

##### 2. PRODUCTION (FARMING):

- **Activities:** Growing the crops.
- **Opportunity:** This is the traditional role. The opportunity is to specialize in "high-value" or "niche" crops (e.g., mushrooms, strawberries, herbs, organic vegetables) instead of common crops.

##### 3. AGGREGATION & COLLECTION:

- **Activities:** Buying produce from many small farmers, grading it by size and quality, and packaging it into large, uniform batches.
- **Opportunity:** Become an "aggregator" or "middle-man." You solve the problem of small, inconsistent supplies.

#### 4. PROCESSING (VALUE ADDITION):

- **Activities:** This is the most important stage. It is changing the *form* of the raw product to increase its shelf life and value.
- **Examples:**
  - Mangoes -> Dried mango slices, Mango juice, Mango jam.
  - Tomatoes -> Tomato paste, Ketchup, Canned tomatoes.
  - Cassava -> Garri, High-Quality Cassava Flour (HQCF), Starch.
  - Peppers -> Dried, ground pepper (spice).
  - Oranges -> Orange juice, Orange oil (from the peel).
- **Opportunity:** Start a small-scale processing business. Even simple solar dryers for fruit can add huge value.

#### 5. MARKETING & DISTRIBUTION:

- **Activities:** Transport (especially "cold chain" logistics), branding, advertising, and sales.
- **Opportunity:** A "cold chain" logistics business (renting refrigerated vans) is a massive, untapped opportunity in Nigeria to reduce post-harvest losses.

#### 6. RETAIL:

- **Activities:** Selling directly to the consumer.
- **Opportunity:** A modern, clean fruit and vegetable store. A subscription "veg box" service that delivers fresh produce to homes.

#### Evaluation questions

1. What is a "value chain"?
2. What is the difference between "Farm Gate Price" and "Retail Price"?
3. What is "value addition"?
4. Give two examples of processing (value addition) for a tomato.
5. What is "cold chain" logistics?

### Assignment questions

1. Draw a simple diagram of the value chain for a mango, showing the five main stages.
2. List three distinct agribusiness *opportunities* (jobs) that are *not* "farming."
3. Why is "aggregation" an important business in a country with many small-scale farmers?
4. Why does processing (e.g., drying mangoes) "add value"? What problem does it solve for the consumer?
5. Choose a horticultural crop (e.g., orange, pepper, pineapple) and list three value-added products that can be made from it.



## WEEK 4: FARM MANAGEMENT

### TOPIC: PLANNING, BUDGETING, RECORD KEEPING

#### Introduction

A farm is a business. To be successful, it must be managed like any other business. Farm management is the science of making decisions to achieve the farmer's goals (usually, to make a profit). The three key tools for this are planning, budgeting, and record-keeping.

#### A. Farm planning

- **Definition:** This is the process of deciding *in advance* what to do, how to do it, and when to do it. It is the "map" for the farm's future.
- **Key planning decisions:**
  - **What to produce?** (e.g., Maize or yams? Tomatoes or peppers? Based on market demand and land suitability).
  - **How much to produce?** (e.g., 1 hectare or 5 hectares? Based on available capital and labour).
  - **How to produce?** (e.g., Organic or conventional? Manual labour or mechanization? Rain-fed or irrigation?).
- **The Farm Plan:** A written document that outlines all activities for the season (e.g., "Week 1: Clear land. Week 2: Buy seeds. Week 4: Plant...").

#### B. Farm budgeting

- **Definition:** A budget is a *financial plan* for the farm. It is an *estimate* of all expected income and all expected expenses over a period of time (e.g., one growing season).
- **Importance:**
  - It predicts if the farm will be profitable or not.
  - It is essential for getting a bank loan.
  - It helps control spending.
- **Components of a budget:**

1. **Revenue (Income):** Expected Yield (e.g., 200 baskets of tomatoes) x Expected Price (e.g., ₦5,000/basket) = ₦1,000,000

2. **Expenses (Costs):**

- **Variable costs:** Costs that *change* with the size of production (e.g., seeds, fertilizer, daily-paid labour, transport, baskets).
- **Fixed costs:** Costs that *do not change* with production (e.g., rent, manager's salary, depreciation on tools, insurance).

• **Profit (or Loss) = Total Revenue - Total Expenses**

### C. Farm record keeping

- **Definition:** This is the *most important* part of farm management. It is the *history book* of the farm. It is the systematic writing down of all activities, observations, and financial transactions.
- **"You cannot manage what you do not measure."**
- **Types of farm records:**
  1. **Production records:** Date of planting, date of weeding, fertilizer application rates, pest outbreaks, rainfall data, date of harvest, final yield (kg).
  2. **Financial records:** (The most critical). A cash book or ledger that lists:
    - *All expenses:* "March 5: Bought 2 bags NPK - ₦30,000."
    - *All income:* "June 10: Sold 5 baskets tomato - ₦25,000."
  3. **Inventory records:** A list of all assets (tools, machines, land) and their value.
- **Importance:** Records are used to create the *next* budget (they replace "estimates" with "facts"), secure loans, and make better management decisions.

### Evaluation questions

1. Define "farm management."
2. What is the difference between a "fixed cost" and a "variable cost"? Give one example of each.
3. Why is a farm budget important? (List two reasons).
4. What is a "production record"? Give two examples of data you would find in one.

5. What is the relationship between "record keeping" (past) and "budgeting" (future)?

### Assignment questions

1. Explain the difference between farm *planning* and farm *budgeting*.
2. List three types of farm records and explain the importance of *each* one.
3. Why would a bank *refuse* to give a loan to a farmer who does not have farm records?
4. What does the phrase "You cannot manage what you do not measure" mean in the context of farm management?
5. **Practical:** Create a simple, one-season "farm budget" for a 1-hectare maize farm. (Estimate your revenue and list at least 3 variable costs and 2 fixed costs).





## WEEK 5: MIDTERM EXAMINATION

### Topic: Examination

This week is for the midterm test, which will cover all content from Weeks 1-4.

- **Topics:** Export Crops (Cocoa, Cashew, Kola), Global Trends (Agribusiness, Organic Certification), Value Chain, and Farm Management (Planning, Budgeting, Records).
- **Format:** A mix of objective, short-answer, and essay questions.



## WEEK 6: AGRICULTURAL FINANCE

### TOPIC: SOURCES OF CREDIT, CHALLENGES

#### Introduction

Farming is a seasonal business. A farmer invests money in land, seeds, and labour at the beginning of the season but does not get paid until the harvest, months later. This "cash flow gap" requires **agricultural finance** (money to run the farm) and **credit** (borrowed money).

#### A. Key terms

- **Finance:** The management of money, including saving, lending, and investing.
- **Credit:** Receiving money now that you promise to pay back in the future, almost always with an "interest" (a fee for borrowing).
- **Collateral:** An asset (like a house, car, or land document) that a borrower promises to give the lender if they fail to pay back the loan.

#### B. Sources of agricultural credit

##### 1. Formal sources (Official, regulated, and documented):

- **Commercial banks:** (e.g., UBA, First Bank, Zenith Bank). They offer loans to large, established agribusinesses but are often reluctant to lend to small-scale farmers.
- **Microfinance banks:** (e.g., LAPO, Grooming Centre). They specialize in small loans to small businesses and are better suited to farmers, but often have higher interest rates.
- **Government banks / agencies:**
  - **Bank of Agriculture (BOA):** The main government bank for farmers, offering low-interest loans.
  - **Central Bank of Nigeria (CBN):** Runs many "intervention" programmes, like the **Anchor Borrowers' Programme (ABP)**.

- **Bank of Industry (BOI):** Funds the processing and industrial side of agriculture.

## 2. **Informal sources (Unofficial, not regulated, based on trust):**

- **Cooperatives & "Esusu" (Ajo):** Groups of farmers pool their savings together and give loans to members. This is a very common and effective source.
- **Family and friends:** The most common source for very small loans. Interest-free, but not reliable.
- **Moneylenders:** Private individuals who lend money at *extremely* high interest rates. They are easy to get but very dangerous and can lead to a cycle of debt.

## C. Challenges in agricultural finance

Why is it so hard for farmers to get loans in Nigeria?

1. **High risk:** Farming is risky (drought, pests, floods, price changes). Banks are afraid the farmer will lose the crop and be unable to repay.
2. **Lack of collateral:** Most rural farmers do not have the legal "Certificate of Occupancy" (C of O) for their land, so they have no collateral to offer the bank.
3. **Lack of farm records:** (Link to Week 4). Without a budget and financial records, the farmer cannot *prove* to the bank that their farm is a profitable, well-managed business.
4. **High interest rates:** Commercial loans are often too expensive (e.g., 20-30% interest), making it impossible to profit.
5. **Loan diversion:** Some farmers use the "farm loan" for personal needs (e.g., school fees, medical bills) and then have no money for the farm.

## Evaluation questions

1. What is "credit"?
2. What is "collateral"?
3. List two "formal" sources and two "informal" sources of agricultural credit.
4. Why is the Bank of Agriculture (BOA) a better option for a farmer than a commercial bank?
5. What is the main danger of borrowing from an informal "moneylender"?

### Assignment questions

1. What is the "cash flow gap" in farming, and why does it make credit so important?
2. Why is "lack of collateral" the single biggest challenge for small-scale farmers seeking loans?
3. How does good "farm record keeping" (from Week 4) help solve the problem of "agricultural finance"?
4. What is a "farmers' cooperative," and how does it help members get credit?
5. What are the *risks* for a *bank* when lending to farmers? (List 3).



## WEEK 7: MIDTERM BREAK



## WEEK 8: AGRICULTURAL POLICIES IN NIGERIA

### TOPIC: GOVERNMENT PROGRAMMES AND IMPACTS

#### Introduction

An agricultural policy is a set of goals, laws, and actions (programmes) created by the government to influence and manage the country's agricultural sector. These policies change with different governments and have had varying degrees of success.

#### A. General goals of agricultural policy

Most government policies for agriculture aim to:

1. **Achieve food security:** (Ensure a stable, affordable supply of food for the population).
2. **Generate foreign exchange:** (Boost the export of cash crops).
3. **Create rural employment:** (Reduce poverty by creating jobs on farms and in agribusiness).
4. **Provide raw materials:** (Supply local industries, e.g., fruit juice factories).

#### B. Examples of past and present programmes

1. **Operation Feed the Nation (OFN) (1976):** Launched by the Obasanjo military government. It was a mass campaign to encourage *all* Nigerians (students, civil servants) to get involved in farming.
  - **Impact:** A well-intentioned "social" campaign, but it failed to increase food production significantly because it lacked technical support, inputs, and proper planning.
2. **Green Revolution Programme (1979):** Launched by the Shagari government. The goal was to boost production using modern inputs (high-yield seeds, fertilizers, tractors).
  - **Impact:** It focused on large-scale, mechanized farms, often ignoring small-scale farmers. It was largely unsuccessful due to poor implementation and corruption.

3. **Directorate of Food, Roads and Rural Infrastructure (DFRRI) (1986):** Launched by the Babangida government. The idea was to build rural roads, provide water, and electrify villages, which would *indirectly* boost agriculture.
  - **Impact:** A very good idea in theory, as it tackled a key problem (transport). It failed due to massive corruption and lack of maintenance.
4. **Agricultural Transformation Agenda (ATA) (2011):** Launched by the Jonathan government. This was a major policy shift to treat "Agriculture as a Business."
  - **Key programme:** The "E-Wallet" (Growth Enhancement Support - GES) scheme. This bypassed corrupt middlemen by sending farmers vouchers for fertilizer and seeds *directly to their mobile phones*.
  - **Impact:** It was very successful at cleaning up the fertilizer subsidy system.
5. **Anchor Borrowers' Programme (ABP) (2015):** Launched by the Buhari government (run by CBN).
  - **How it works:** It links a large processor (the "Anchor") with thousands of small farmers. The Anchor gets a large loan from the CBN, gives it to the farmers as "inputs" (seeds, fertilizer - not cash), and then guarantees to buy the entire harvest from the farmers.
  - **Impact:** Massively increased the production of rice in Nigeria, but has faced major challenges with farmers not repaying their loans.

### Evaluation questions

1. What is an "agricultural policy"?
2. List three main *goals* of agricultural policy in Nigeria.
3. What was "Operation Feed the Nation (OFN)"?
4. What was the "E-Wallet" system, and what problem did it solve?
5. Describe how the "Anchor Borrowers' Programme" works.

### Assignment questions

1. What is "policy inconsistency" (i.e., new governments abandoning old programmes), and how has it affected Nigerian agriculture?

2. What was the main *flaw* of the "Green Revolution Programme" in the 1970s?
3. Why did the "DFRRI" programme fail, even though the idea was good?
4. In your opinion, which of the programmes listed was the *most* successful? Justify your answer.
5. The ABP has been praised for boosting rice production but criticized for poor loan repayment. Why do you think many farmers failed to repay their loans?





## WEEK 9: AGRICULTURAL MARKETING BOARDS

### TOPIC: FUNCTIONS, ROLES, LIMITATIONS

#### Introduction

From the 1940s until the 1980s, the government managed the export of cash crops through powerful agencies called "Marketing Boards." They had a total monopoly on the purchase and sale of crops like cocoa, palm oil, and groundnuts.

#### A. Structure of the marketing boards (e.g., The Cocoa Board)

- **Monopoly:** The Board was the *only* legal buyer of cocoa from farmers. It was illegal for a farmer to sell cocoa to anyone else.
- **Structure:** The Board set a "guaranteed price" at the start of each season. It used "Licensed Buying Agents" (LBAs) to go to villages and buy the cocoa, which was then graded and exported by the Board.

#### B. Intended functions and roles

1. **Price stabilization:** This was the main idea. The world price of cocoa changes every day. The Board would *protect* farmers from low prices by guaranteeing a stable price for the whole season. In "good" years (high world price), the board would pay the farmer less and *save* the profit. In "bad" years (low world price), it would use the savings to pay the farmer *more*.
2. **Quality control:** The Boards set and enforced quality standards. They inspected and graded all produce, ensuring that Nigerian cocoa had a good reputation internationally.
3. **Revenue for development:** The "profits" made by the Boards (by paying farmers less than the world price) were a major source of revenue for the government, funding schools, roads, and universities in the 1950s and 60s.
4. **Funding research:** The Boards funded research institutes like the Cocoa Research Institute of Nigeria (CRIN) and the Nigerian Institute for Oil Palm Research (NIFOR).

#### C. Limitations and failure

The system worked well at first, but by the 1970s and 80s, it had failed completely.

1. **Exploitation (Low Prices):** The "tax" on farmers became exploitation. The Boards paid farmers a *tiny* fraction of the world price. Farmers became very poor.
2. **Corruption and inefficiency:** The Boards became bloated, corrupt government agencies that wasted money.
3. **Disincentive to produce:** Because prices were so low, farmers stopped taking care of their farms. Many abandoned their cocoa farms to look for other work.
4. **Smuggling:** Farmers began smuggling their cocoa to neighbouring countries (like Benin Republic) where they could get a much higher price from private buyers.

**Abolition:** Due to these failures, the Marketing Boards were abolished in 1986 as part of the Structural Adjustment Programme (SAP).

### Evaluation questions

1. What is a "monopoly"?
2. What was the *main* intended function of the Marketing Boards?
3. What is "price stabilization"?
4. List two positive roles the Marketing Boards played.
5. List two major *limitations* (failures) that led to their abolition.

### Assignment questions

1. Compare the "Marketing Board" era with the "Liberalized Market" (today), where any private company can buy and export cocoa. What are the pros and cons of *each* system?
2. How did the Marketing Boards contribute to the development of Nigeria in the 1950s and 60s?
3. Explain how the low prices set by the Boards created a "disincentive to produce."
4. Why did "smuggling" become a major problem for the Marketing Boards?
5. Do you think the government should bring back Marketing Boards? Give one reason for and one reason against.

## WEEK 10: PROJECT PLANNING

### TOPIC: PREPARING FOR FINAL PROJECT

#### Introduction

Your final year project is the most important practical work you will do in SS3. It is a chance to apply the knowledge you have learned in horticulture and agribusiness. It can be a practical project, a research study, or a business plan. This week, we learn how to plan it.

#### A. Choosing a project topic

A good topic must be "SMART":

- **Specific:** Clear and well-defined. (Bad: "My project is on farming." Good: "My project is on the effect of NPK on tomato yield.")
- **Measurable:** You must be able to measure your results. (e.g., "I will measure the *height* and *weight* of the tomatoes.")
- **Achievable:** You must have the skills, time, and money to do it. (Bad: "I will build a tractor." Good: "I will build a small rabbit hutch.")
- **Relevant:** It must be related to Horticulture and Crop Production.
- **Time-bound:** It must be possible to finish it this term.

#### B. Project ideas

- **Horticulture (Practical):** "Setting up a 10-plant Deep Water Culture (DWC) hydroponic system."
- **Horticulture (Experiment):** "Comparing the growth of maize using poultry manure vs. NPK 15:15:15."
- **Value Addition (Practical):** "Producing and packaging 1kg of solar-dried mango slices."
- **Agribusiness (Research):** "A survey of post-harvest losses of tomatoes at Ojuwoye Market."
- **Business Plan (Management):** "A complete business plan and budget for a 1-acre cucumber farm."

### C. The project report (The 5-chapter format)

All projects must be written up as a formal report.

- **Cover page & Table of contents**
- **CHAPTER 1: INTRODUCTION**
  - **Background:** What is the topic? Why is it important?
  - **Problem statement:** What problem are you trying to solve?
  - **Aim and objectives:** What is the main goal (Aim)? What are the specific steps (Objectives)?
- **CHAPTER 2: LITERATURE REVIEW**
  - What do books, the internet, and past projects *already* say about your topic? This shows you have done your research.
- **CHAPTER 3: METHODOLOGY**
  - The "How-To" chapter. *Exactly* what you did, step-by-step.
  - What materials did you use? What was your experimental design? How did you collect your data?
- **CHAPTER 4: RESULTS AND DISCUSSION**
  - **Results:** Show your data (tables, charts, photos). *What did you find?*
  - **Discussion:** *What does it mean?* Why did it happen?
- **CHAPTER 5: CONCLUSION AND RECOMMENDATIONS**
  - **Conclusion:** A short summary of your findings. Did you achieve your aim?
  - **Recommendations:** What would you suggest based on your project? What are the "next steps"?
- **References**

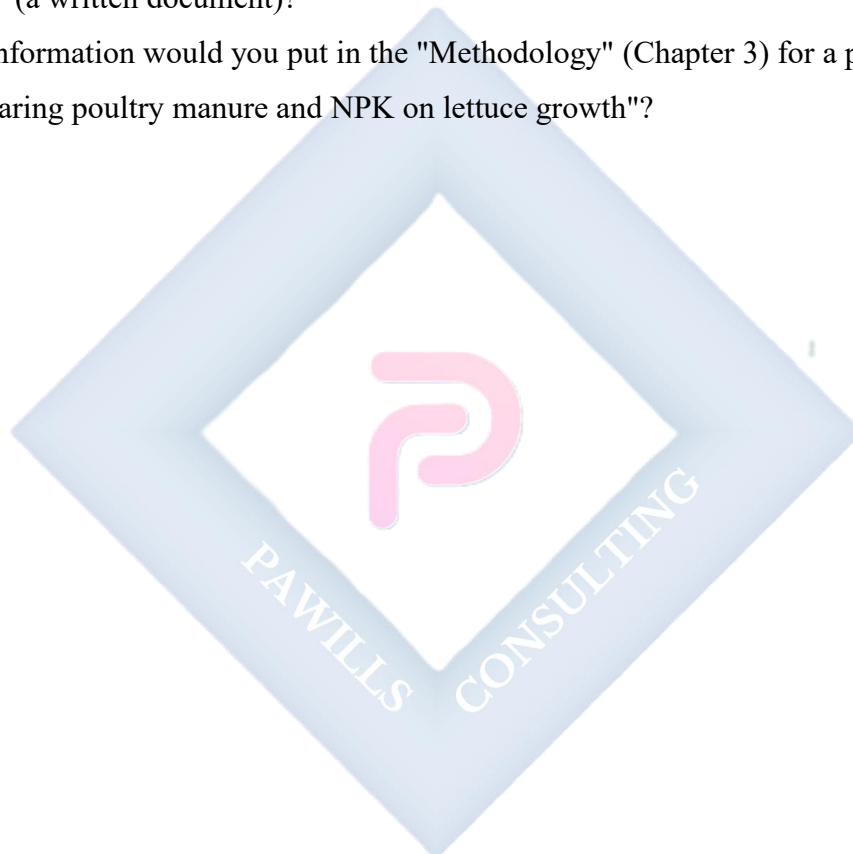
### Evaluation questions

1. What does the "A" in the "SMART" acronym for project topics stand for?
2. What is the "Literature Review" (Chapter 2) of a project?
3. Which chapter is the "How-To" guide of your project?
4. What is the difference between "Results" (data) and "Discussion" (meaning)?

5. What is the "Problem Statement" in Chapter 1?

### **Assignment questions**

1. Write *two* "SMART" project topics that you would be interested in doing.
2. Choose one of your topics and write the "Aim" and three "Objectives" for it.
3. Why is a "Literature Review" important? (What does it prevent you from doing?)
4. What is the difference between a "Project" (a practical/research activity) and a "Project Report" (a written document)?
5. What information would you put in the "Methodology" (Chapter 3) for a project titled "Comparing poultry manure and NPK on lettuce growth"?



## **WEEK 11-12: REVISION & EXAMINATION**

- **WEEK 11:** Comprehensive review of all topics from the term. Focus on the economic and business aspects of agriculture (Value Chain, Finance, Management) and the role of government (Policies, Marketing Boards).
- **WEEK 12:** Final Term Examination.



## SS3 HORTICULTURE AND CROP PRODUCTION LESSON NOTES

### THIRD TERM

#### WEEK 1: FINAL YEAR PROJECT SUPERVISION

#### TOPIC: PROPOSAL WRITING

##### Introduction

This term is the most critical of your entire study, as it culminates in your Final Year Project. This project is a practical application of all the knowledge you have acquired in SS1, SS2, and SS3. It is your chance to function as a real horticulturalist, researcher, or agribusiness professional. The foundation of a good project is a great proposal.

##### A. What is a project proposal?

A project proposal is a formal, written document that outlines your intended project. It is a "roadmap" that explains *what* you are going to do, *why* it is important, and *how* you are going to do it. It must be approved by your supervisor before you begin.

##### Purpose of the proposal:

1. **To force you to think clearly:** Writing down your idea helps you identify potential problems and refine your topic.
2. **To get approval:** It proves to your teacher that your project is relevant, achievable, and not too simple or too complex.
3. **To guide your work:** It becomes the "contract" for your project and the outline for your Chapter 1.

##### B. Choosing a "SMART" project topic

Your topic must be **SMART**:

- **Specific:** Clear and focused.

- *Bad*: "A project on tomatoes."
- *Good*: "The effect of poultry manure on the growth and yield of Roma tomatoes."
- **Measurable**: You must be able to collect data.
  - *Good*: "I will measure the plant height (cm) and final fruit weight (kg)."
- **Achievable**: You must have the resources (time, money, materials) to do it.
  - *Bad*: "Building a high-tech automated greenhouse."
  - *Good*: "Constructing a 1m x 2m drip irrigation system for a vegetable bed."
- **Relevant**: It must be related to horticulture and crop production.
- **Time-bound**: It must be possible to complete within the 4-6 weeks of this term.

### C. Example horticulture project ideas

1. **Experimental project**: "The comparative effect of NPK 15:15:15 and poultry manure on the yield of leafy green (Ugu)."
2. **Soilless culture project**: "Setting up a small-scale (10-plant) Deep Water Culture (DWC) hydroponic system for growing lettuce."
3. **Value addition project**: "The production, packaging, and shelf-life analysis of solar-dried mango slices."
4. **Agribusiness/Survey project**: "A market survey of post-harvest losses of tomatoes at [Your Local Market]."
5. **Design project**: "The design and construction of a low-cost, gravity-fed drip irrigation system for a home garden."

### D. Key components of a project proposal

Your proposal should be a short document (2-3 pages) with these sections:

1. **Project title**: Clear, descriptive, and concise.
2. **Introduction / Background of the study**: (1-2 paragraphs) What is the problem? Why is this project important? (e.g., "Post-harvest losses of tomatoes are a major problem... This project will explore solar drying as a solution...").
3. **Aim and objectives**:



- **Aim:** The *one* main goal of your project. (e.g., "The aim is to determine the best fertilizer for growing Ugu.").
- **Objectives:** The 3-5 specific *steps* you will take to achieve the aim. (e.g., "1. To set up three experimental plots. 2. To apply NPK, poultry manure, and a control. 3. To collect data on plant height... 4. To compare the final yield.").

#### 4. **Methodology:**

- How will you do the work? What tools, materials, and processes will you use?
- (e.g., "The project will use 3 plots, 20 seedlings, a ruler for measurement, and a scale for final yield...").

#### 5. **Scope and limitations:**

- What are the boundaries? (e.g., "This project will only test two fertilizers and will last for 6 weeks.").

#### 6. **Expected outcome:**

- What will you have at the end? (e.g., "A 5-chapter project report and a recommendation for the best fertilizer.").

### **Evaluation questions**

1. What does the "A" in the "SMART" acronym stand for, and why is it important?
2. What is the main difference between the "Aim" and the "Objectives" of a project?
3. What is the "Methodology" section of a proposal?
4. Why is a "Background of the Study" necessary?
5. What is the purpose of a project proposal?

### **Assignment questions**

1. Write *two* "SMART" project titles for a horticulture project you are interested in.
2. Choose one of your project titles. Write a clear "Aim" and three "Objectives" for it.
3. List the top 10 "Tools and Materials" you would need for the project you chose in question 2. (This will become your Methodology).
4. Write a 1-paragraph "Introduction" for your chosen project, explaining the "problem" it is trying to solve.

5. What is the "Scope" of your project? (What will you *not* be doing?).



## WEEK 2: PROJECT EXECUTION

### TOPIC: PRACTICAL CROP/HORTICULTURAL PROJECTS

#### Introduction

With your proposal approved, the practical work begins. This is the "hands-on" phase where you set up your experiment, plant your crops, or build your system. The single most important part of this phase is **documentation**. If you don't write it down, it didn't happen.

#### A. Setting up the project

1. **Site selection:** Choose a good location (e.g., school farm, home garden) with good sunlight and access to water.
2. **Land preparation:** Clear the land, till the soil, and create your beds or ridges.
3. **Experimental design (for experimental projects):**
  - You *must* have a **Control Group**.
  - *Example:* For a fertilizer project, you need three plots:
    - **Plot A (Control):** Soil only, no fertilizer.
    - **Plot B (Test 1):** Soil + NPK fertilizer.
    - **Plot C (Test 2):** Soil + Poultry Manure.
  - This is the only way to *prove* that the fertilizer caused the change.
4. **Procurement:** Gather all your materials (seeds, seedlings, manure, tools, hydroponic nutrients, etc.).

#### B. The project logbook (Your "Work Diary")

This is the most critical document you will create. Your entire project report (especially Chapter 3 and 4) will be written from this logbook.

- You must use a dedicated notebook.
- You must make an entry *every time* you work on your project.
- **Each entry must include:**
  - **Date:** e.g., "October 14, 2025"

- **Task performed:** "Planted 20 tomato seedlings. Watered all plots."
- **Observations:** "Noticed 3 seedlings in Plot B are wilting. Saw two grasshoppers on Plot A."
- **Financials:** "Bought 1 bag of manure - ₦1,500."

### C. Data collection

You must collect *measurable* data to analyse in Chapter 4.

- **What to measure:**
  - **Germination:** "Number of seeds germinated" (e.g., 8 out of 10).
  - **Growth:** "Plant height (cm)" (measure this weekly). "Number of leaves."
  - **Flowering:** "Date of first flowering." "Number of flowers."
  - **Yield:** "Number of fruits." "Total weight of fruits (kg)."

### Evaluation questions

1. What is a "Control Group," and why is it essential for an experimental project?
2. What is a "Project Logbook"?
3. List 4 pieces of information that must be in every logbook entry.
4. What is the difference between "setting up" and "data collection"?
5. List 3 *measurable* data points you can collect from a tomato plant.

### Assignment questions

1. Write your *first* two logbook entries for your project (e.g., "Land Preparation" and "Planting").
2. Take a "Before" photo of your project site or materials.
3. Create a simple "Data Collection Sheet" (a table) that you will use to record your weekly measurements (e.g., Date, Plot A Height, Plot B Height, Plot C Height, Notes).
4. What is the biggest *safety risk* in your specific project? (e.g., "Chemical fertilizer burns," "Sharp cutlass," "Electric pump").

5. What is your "Experimental Design"? (Describe your "Control" plot and your "Test" plots).



## WEEK 3: PROJECT EXECUTION II

### TOPIC: FIELD WORK AND DEMONSTRATIONS

#### Introduction

This week is about continuing the day-to-day management of your project and dealing with the inevitable challenges. A project rarely goes perfectly. How you observe, record, and solve problems is a key part of the process.

#### A. Ongoing project management

1. **Cultural practices:** You must perform all necessary tasks.
  - **Weeding:** Control weeds, as they compete for nutrients and light.
  - **Watering:** Ensure a consistent water supply, especially for soilless cultures.
  - **Fertilization:** Apply your treatments (e.g., NPK, manure) as per your methodology.
  - **Pest/Disease control:**
    - **Observe:** Check your plants *daily* for signs of pests (e.g., holes in leaves) or disease (e.g., spots, wilt).
    - **Record:** Take photos and notes in your logbook.
    - **Control:** Use appropriate methods (e.g., hand-picking pests, applying neem oil or appropriate pesticides).

#### B. Troubleshooting

- **Problem:** "My hydroponic plants are wilting."
  - *Troubleshoot:* Is the pump working? Is the nutrient solution level too low? Is it too hot?
- **Problem:** "My seedlings are being eaten."
  - *Troubleshoot:* What is the pest? (e.g., Snails, grasshoppers, birds). Apply the correct control (e.g., netting, organic pellets, hand-picking).

#### C. Photographic evidence

You *must* take photos at every stage. They are essential for your final report and presentation.

- Photo 1: Before (Land preparation).
- Photo 2: Planting.
- Photo 3: Mid-Growth (e.g., Week 3).
- Photo 4: Any "Problem" (e.g., pest damage, wilted plants).
- Photo 5: Harvest (The final product).

#### **D. Milestone demonstration**

- At the end of this week, you must be prepared to give your supervisor a 2-minute "progress report."
- You must show:
  1. Your **physical project** (the plants, the system).
  2. Your **Project Logbook** (up-to-date).

#### **Evaluation questions**

1. List three ongoing "cultural practices" you might perform this week.
2. Why is it important to take photos of "problems" (like pest damage)?
3. What two items must you show your supervisor in a "milestone demonstration"?
4. Why is it important to check your project daily?
5. What is the first step in troubleshooting a problem?

#### **Assignment questions**

1. Write a logbook entry for a "problem" you faced this week (or a potential problem). (e.g., "Date: ... Task: Inspected plants. Observation: Found 5 aphids on... Action: Sprayed with...").
2. Take a "mid-growth" photo of your project.
3. What is the *current status* of your project? (e.g., "Germination complete," "Waiting for flowering," "System built").
4. What is the *next* single step you need to take?

5. What is one *cost* you have had this week? Record it in your financial log.





## WEEK 4: PROJECT REPORT WRITING

### TOPIC: DOCUMENTATION AND ANALYSIS

#### Introduction

The project is not finished until the report is written. A perfectly-grown crop with no report is a failure. The report proves you did the work and, more importantly, *you understood the work*.

This is the single most important document of your SS3.

#### A. The 5-chapter project structure (WAEC/NECO Standard)

Your final report must follow this format. Use your logbook and proposal to write it.

- **Cover Page:** Your Title, Your Name, School, Date.
- **Table of Contents.**

#### 1. CHAPTER 1: INTRODUCTION

- (This is just an expansion of your proposal).
- Background of the Study
- Problem Statement
- Aim and Objectives
- Scope and Limitations
- Significance of the Project (Why it matters)

#### 2. CHAPTER 2: LITERATURE REVIEW

- This is the "theory" behind your project.
- What do books, the internet, and past projects *already* say about your topic?
- (e.g., "What is hydroponics?", "What nutrients do tomatoes need?", "What is NPK fertilizer?").

#### 3. CHAPTER 3: METHODOLOGY

- This is the "How-To" chapter. It is written in the **PAST TENSE**.
- Use your logbook to write this!
- List all Tools Used, Materials Used.

- Describe your *process* step-by-step. (e.g., "The land *was* cleared... Three plots *were* marked... The seeds *were* planted on [Date]...").

#### 4. CHAPTER 4: RESULTS AND ANALYSIS

- **This is the most important chapter.**
- **Results:** This is your *data*. **Do not** explain. Just show.
  - Use tables (e.g., "Table 1: Weekly Plant Height (cm)").
  - Use charts (e.g., A bar chart comparing the final yield of Plot A, B, and C).
  - Show your photos.
- **Analysis (Discussion):** This is the *meaning*.
  - *What do your results mean?*
  - (e.g., "As shown in the bar chart, Plot C (Poultry Manure) had the highest yield at 5.2kg, which was 30% more than the control plot. This shows that poultry manure is highly effective...").

#### 5. CHAPTER 5: CONCLUSION AND RECOMMENDATIONS

- **Conclusion:** (1 paragraph) A summary of your findings. Did you achieve your aim? (e.g., "This project found that poultry manure produced the highest yield...").
- **Recommendations:** (Bulleted list) What would you recommend based on your findings? (e.g., "1. Farmers should use poultry manure... 2. Future projects should test different *amounts* of manure...").

#### Evaluation questions

1. What are the five mandatory chapters of a final year project?
2. Which chapter is written in the past tense and is based on your logbook?
3. What is the "Literature Review" (Chapter 2) for?
4. What is the critical difference between "Results" and "Analysis" in Chapter 4?
5. What two things go in Chapter 5?

#### Assignment questions

1. **PRACTICAL:** Write your full Chapter 1 (you should have most of this from your proposal).

2. **PRACTICAL:** Write your full Chapter 3 (use your logbook and list all your tools, materials, and steps in the *past tense*).
3. **PRACTICAL:** Create one "Table" and one "Bar Chart" for your results (even if you have to use practice data).
4. **PRACTICAL:** Write your full Chapter 5 (Conclusion and Recommendations).
5. Insert your "Before" and "After" photos into a document and write a "caption" for each one, explaining what it shows.



## **WEEK 5: MIDTERM EXAMINATION**

### **TOPIC: PROJECT SUBMISSION AND MOCK DEFENSE**

#### **Content**

This is the submission deadline for the first draft of your project report and a "mock defense" (a practice presentation).

- **Part 1: Report submission:** You will submit your full, 5-chapter report for review.
- **Part 2: Practical demonstration:** You will demonstrate your project (the plants, the harvest, the product) to your teacher.
- **Part 3: Mock defense:** You will give a 5-minute verbal presentation to your teacher, explaining what you did and what you learned.

#### **Evaluation questions**

1. What are the three parts of your midterm examination?
2. What is the purpose of a "mock defense"?
3. What is the most important thing to show in your "practical demonstration"?
4. What chapter of your report details your findings and photos?
5. What chapter details your step-by-step process?

#### **Assignment questions**

1. Check your report for spelling and grammar.
2. Practice your 5-minute presentation speech.
3. Prepare your project/harvest/product for the demonstration.
4. Anticipate one question your teacher might ask you about your project, and write down the answer.
5. Review the "feedback" from your supervisor on your report and make the necessary corrections.

## WEEK 6: PRESENTATION SKILLS

### TOPIC: ORAL DEFENSE, ICT USE

#### Introduction

Your "oral defense" is the final part of your project. You will stand and present your work. Your goal is to prove to the examiners that *you* did the work, *you* understood the theory, and *you* analysed the results.

#### A. Creating your presentation (ICT use)

Use simple ICT tools like PowerPoint or Google Slides. Keep it simple, visual, and professional.

**Do not write long paragraphs of text.**

- **Slide 1: Title Page:** Project Title, Your Name, School.
- **Slide 2: Introduction:** (1-2 sentences) What was the problem? What was your aim?
- **Slide 3: Methodology:** (Use *pictures* of your tools and you working, not just text).
- **Slide 4: Results! (The "Hero" Slide):** Show your best "Before" and "After" photos. Show your bar chart or results table.
- **Slide 5: Analysis:** (1-2 bullet points) What does your data *mean*?
- **Slide 6: Conclusion & Recommendations:** What did you learn?
- **Slide 7: Thank You & Questions:**

#### B. "The 10-20-30 Rule"

A good guide for presentations:

- **10** Slides (Maximum)
- **20** Minutes (Maximum, your defense will be 5-10 mins)
- **30**-point font size (or larger). This forces you to use less text.

#### C. Speaking and presenting

1. **Do not read the slides!** The slides are for your audience; your *speech* is from your head.

2. **Make eye contact:** Look at your teacher and classmates.
3. **Speak clearly and confidently:** You are the expert on this topic.
4. **Handle questions:**
  - Listen to the full question.
  - Say "Thank you for your question."
  - Answer it clearly and honestly, using your data.
  - **If you don't know:** Be honest! *Never* lie. Say, "That is an interesting point. My project did not test for that, but it would be a good recommendation for future research."

### Evaluation questions

1. What is the main purpose of an oral defense?
2. What is the *worst* thing to put on a presentation slide?
3. What is the "10-20-30" rule?
4. What should you do if an examiner asks you a question you don't know the answer to?
5. What is the "hero slide" of your presentation?

### Assignment questions

1. **PRACTICAL:** Create your 7-slide presentation.
2. **PRACTICAL:** Practice your 5-minute presentation speech *out loud*.
3. Record your presentation on your phone and watch it. Write down one thing you want to improve.
4. Write down 3 questions you *expect* to be asked.
5. Write down 3 questions you *hope* you are not asked (and then prepare the answers for them).

## **WEEK 7: MIDTERM BREAK**

- Use this break to finalize your project, report, and presentation slides.



## WEEK 8: MOCK EXAMS

### TOPIC: PAST WAEC/NECO AGRICULTURE PAPERS

#### Introduction

With the project finished, our focus now shifts 100% to preparing for your final external examinations. The Agricultural Science exam covers *both* Crop Production and Animal Production. We will use past questions to find your weak areas.

#### A. Exam structure (Typical)

- **Paper 1: Objective (MCQ):** 50 multiple-choice questions (1 hour). Tests a wide range of knowledge (Crops, Animals, Economics).
- **Paper 2: Theory (Essay):** 6 questions, answer 4 (2 hours). Requires in-depth answers.
- **Paper 3: Practical:** 4-5 questions (2 hours). Identification of specimens, tools, pests, and data analysis.

#### B. Mock session: Timed test

- You will be given a full past paper (e.g., WAEC 2022) to complete under exam conditions.
- **Post-Mock review:** We will grade the paper and analyze the most difficult questions. This is *more important* than the test itself. It shows us what to study.

#### C. Key "Hot Topics" from past questions

- **Agriculture systems:** Crop rotation, shifting cultivation.
- **Soil:** Soil profile, soil types, erosion control.
- **Horticulture:** Greenhouse, hydroponics, grafting/budding.
- **Animal:** Ruminant stomach, feed (FCR), diseases.
- **Economics:** Farm records, marketing, credit.

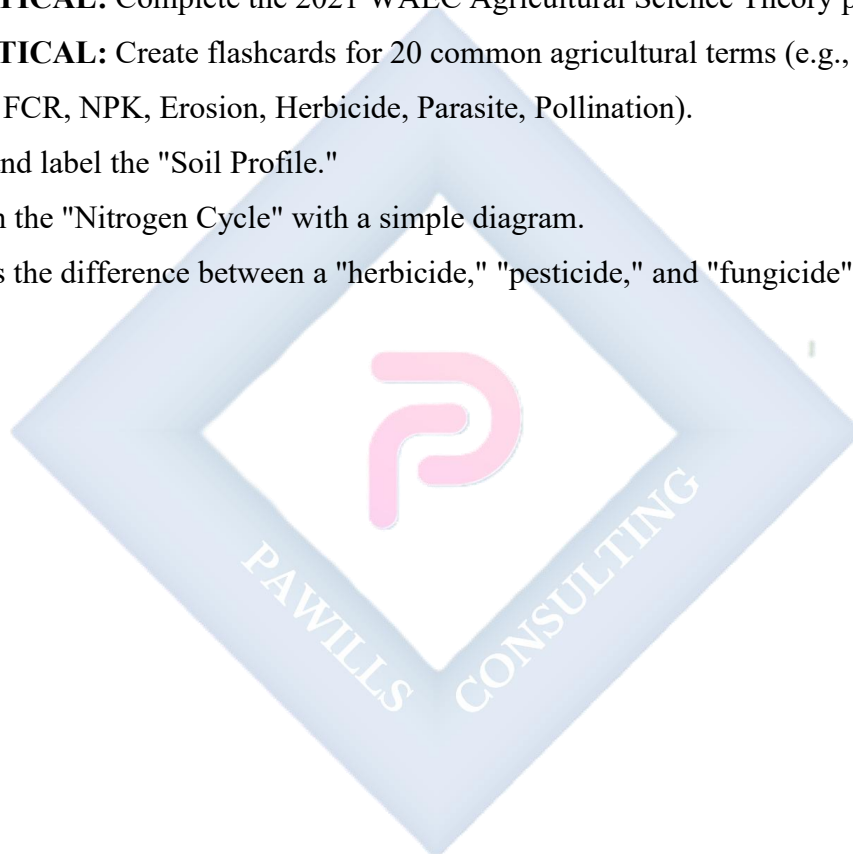
#### Evaluation questions



1. What are the three papers in the external Agriculture exam?
2. What is the purpose of a "post-mock review"?
3. How many questions are in the objective paper?
4. What is "crop rotation"?
5. What are the four chambers of a ruminant stomach?

### **Assignment questions**

1. **PRACTICAL:** Complete the 2021 WAEC Agricultural Science Theory paper at home.
2. **PRACTICAL:** Create flashcards for 20 common agricultural terms (e.g., Hydroponics, Silage, FCR, NPK, Erosion, Herbicide, Parasite, Pollination).
3. Draw and label the "Soil Profile."
4. Explain the "Nitrogen Cycle" with a simple diagram.
5. What is the difference between a "herbicide," "pesticide," and "fungicide"?



## WEEK 9: REVIEW OF PRACTICAL SESSIONS

### TOPIC: HANDS-ON REVISION

#### Introduction

This week is a "boot camp" to refresh your skills for Paper 3 (Practicals). This paper is about *identification* and *application*. We will rotate through stations to review common specimens.

#### A. Station 1: Tools and equipment

- **Identify and state the function:**
  - **Tools:** Secateurs, pruning saw, budding knife, hand trowel, hoe, cutlass, rake, soil auger, wheelbarrow.
  - **Horticultural:** Sprinkler, watering can, grafting tool, shade net.
  - **Surveying:** Ranging pole, measuring tape, Gunter's chain.

#### B. Station 2: Crops and pests

- **Identify:**
  - **Crops:** Maize, Rice, Cassava (tuber/stem), Yam (tuber), Cowpea (seeds), Cocoa (pod), Oil Palm (fruit bunch).
  - **Pests:** (Preserved or photos) Maize weevil, Grasshopper, Aphid, Mealybug, Termite.
  - **Diseases:** (Photos) Leaf spot, Maize smut, Cassava mosaic.

#### C. Station 3: Soil and fertilizers

- **Identify by feel:**
  - **Soil:** Sand (gritty), Clay (sticky), Loam (balanced).
  - **Fertilizers:** NPK (multi-coloured granules), Urea (white, round granules), Muriate of Potash (reddish), SSP (grey), Poultry Manure (organic).

#### D. Station 4: Data analysis

- **Practice calculations:**

- **Plant population:** Area / Spacing
- **Yield analysis:** (e.g., "Analyse the data in the table... which plot was best?").
- **Feed Conversion Ratio (FCR):** Total Feed / Total Weight Gain

### **Evaluation questions**

1. What is the function of a "budding knife"?
2. What is the function of a "soil auger"?
3. How can you tell the difference between NPK and Urea fertilizer by sight?
4. What pest is the main vector for "Cassava Mosaic Disease"?
5. What is the formula for "plant population"?

### **Assignment questions**

1. Draw and label 5 common farm tools.
2. Draw and label 3 common crop pests (e.g., weevil, aphid, grasshopper).
3. How would you identify a "sandy" soil versus a "clay" soil using only your hands?
4. A farmer plants maize on a 2-hectare (20,000 m<sup>2</sup>) field at a spacing of 50cm x 50cm (0.5m x 0.5m). What is the total plant population?
5. A farmer uses 80kg of feed to raise broilers that gain a total of 40kg. What is the Feed Conversion Ratio (FCR)?

## WEEK 10: WAEC/NECO PREPARATORY CLASSES

### TOPIC: LIKELY EXAMINATION TOPICS

#### Introduction

This is our final "cram session." We will focus only on the most valuable, most common, and "guaranteed" topics that appear *every year*.

#### A. "Guaranteed" topics (Review)

1. **Farm records:** (e.g., Production, Inventory, Financial). You must know the *types* and *importance*.
2. **Crop rotation:** You must be able to define it and *draw a 4-year plan*.
3. **Ruminant digestion:** You *must* be able to draw and label the 4-chamber stomach (Rumen, Reticulum, Omasum, Abomasum) and know the function of each.
4. **Farm tools:** Identification and function.
5. **Pests and diseases:** Causative agent, one symptom, and one control method.
6. **Farm management:** Problems of agriculture, roles of government, marketing boards.

#### B. "High-Value" topics (Review)

1. **Soil:** Profile, types, erosion, fertility (NPK, manure).
2. **Cycles:** Nitrogen Cycle, Carbon Cycle, Water Cycle.
3. **Modern tech:** Hydroponics, Greenhouse, AI, Agribusiness value chain.
4. **Horticulture:** Grafting, budding, layering (Definitions and methods).

#### C. Final exam advice

- **Paper 1 (Objectives):** Answer what you know first. Go back for the ones you are unsure about. Do not leave any answer blank.
- **Paper 2 (Theory):** Read the verb. "List" means list. "Explain" or "Describe" means write full sentences. *Always* use diagrams if it helps.

- **Paper 3 (Practical):** Read the instructions carefully. Look at the specimens closely.  
Write clear, concise answers.

### **Evaluation questions**

1. What is the function of the "Rumen"?
2. What is the "Abomasum" often called, and why?
3. Why is "Crop Rotation" practiced?
4. What is the difference between "grafting" and "budding"?
5. List three types of farm records.

### **Assignment questions**

1. Draw a 4-year crop rotation plan for a field, using: Maize (cereal), Beans (legume), Cassava (tuber), and Tomato (vegetable).
2. Draw and label the digestive system of a poultry bird.
3. Draw and label the Nitrogen Cycle.
4. Go through a full WAEC/NECO past question paper. Time yourself.
5. Write a short essay (1 page) on "The problems of agriculture in Nigeria and 5 possible solutions."

## **WEEK 11-12: REVISION & EXAMINATION**

- **WEEK 11:** Final revision of all SS1, SS2, and SS3 work. Focus on practical drills and reviewing theory past questions.
- **WEEK 12:** External Examinations (SSCE) begin.

